



KENYATTA UNIVERSITY

DEPARTMENT OF CIVIL ENGINEERING

ECV 208: STRENGTH OF MATERIALS II – CAT 2

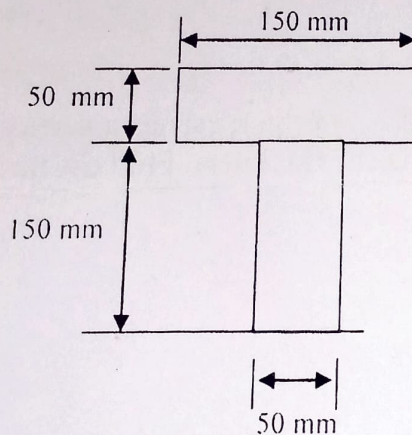
TIME: 1 ½ Hrs

Attempt all questions

Question One (20 Marks)

- a) A 3 m column with the cross section shown is constructed from two pieces of timber. The timbers are nailed together so that they act as a unit. Determine:
- The slenderness ratio.
 - The Euler buckling load given that E for timber is 10.5 Gpa.
 - The axial stress in the column when Euler load is applied.

(7 Marks)



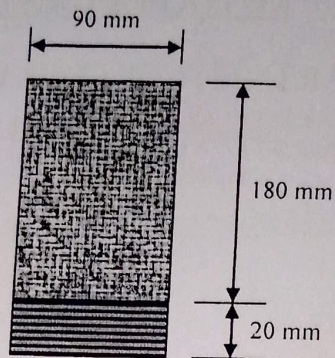
- b) A concrete trapezoidal dam 3 m wide at the top and 10 m wide at the bottom is 24 m high. It contains water on its vertical side. Check the stability of the dam, when it contains water for a depth of 18 m. Take coefficients of friction between the wall and the soil as 0.6 and weight of the concrete as 23.5 kN/m^3 and the allowable compressive strength of concrete as 650 kN/m^2 .

(13 Marks)

Question Two (20 Marks)

- a) A timber beam, 90 mm wide and 180 mm deep is strengthened by a steel plate 90 mm wide and 20 mm thick, screwed at the bottom surface of the timber beam as shown.

(12 Marks)



The modulus of elasticity for timber is 12.5 GPa and for steel is 250 GPa. Knowing that the beam is bent about a horizontal axis by a moment $M=11.6$ kNm, determine the maximum stress in (a) the timber and (b) the steel. Sketch the stress diagram.

- b) A rectangular strut 200 mm wide and 150 mm thick carries a vertical load of 60 kN at an eccentricity of 20 mm in a plane bisecting the thickness. Find the maximum and minimum stress intensities in the section.

(3 Marks)

QUESTION ONE (30 MARKS)

- Define a beam of uniform strength. (3 Marks).
- State five (5) assumptions in the theory of simple bending (5 Marks)
- With illustrations, discuss four different types of end fixtures and the corresponding equivalent column length/height as they relate to Euler's theory for columns. (8 Marks)
- Discuss four different failure modes of a dam and the conditions that must be met to ensure dam stability. (8 Marks)
- Define slenderness ratio. (2 Marks)
- State four (4) assumptions in Euler's theory. (4 Marks)

QUESTION TWO (20 MARKS)

- Find the radius of gyration and the slenderness ratio of a strut made from 4.5 m length of hollow tube 45 mm outer diameter and 35 mm inner diameter. (6 Marks).
- A 2.8 m column with the cross section shown in Figure 1 is constructed from two pieces of timber. The timbers are nailed together so that they act as a unit. Determine:
 - The slenderness ratio
 - The Euler buckling load ($E=12.8 \text{ GPa}$ for timber)
 - The axial stress in the column when Euler load is applied. (14 Marks)

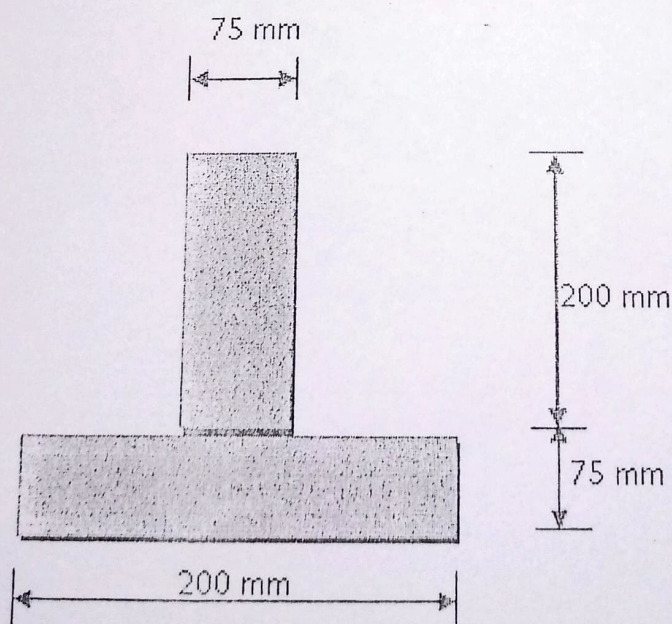


Figure 1

SECTION THREE (20 MARKS)

- a) A masonry trapezoidal dam 1.2 m wide at the top, 4.5 m at its base and 6 m high is retaining water on its vertical face to a height equal to the top of the dam. Determine:
- Total pressure per meter length of the dam
 - Point where the resultant cuts the base
 - Maximum and minimum intensities of stress at the base.
- Take unit weight of water and masonry as 9.81 kN/m^3 and 23.5 kN/m^3 respectively.

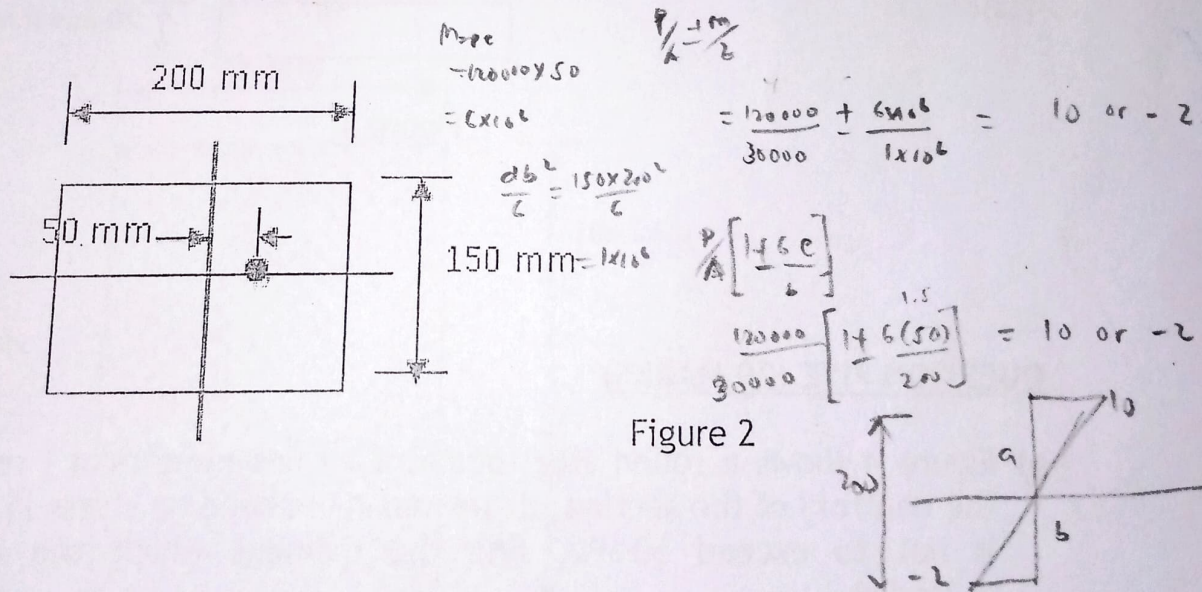
(8 Marks)

- b) A concrete trapezoidal dam 2.5 m wide at the top and 10 m wide at the bottom is 25 m high. It contains water on its vertical side. Check the stability of the dam, when it contains water for a depth of 20 m. Take coefficients of friction between the wall and the soil as 0.6 and weight of the concrete as 24 kN/m^3 . (12 Marks)

QUESTION FOUR (20 MARKS)

- a) A rectangular column 200 mm wide and 150 mm thick is carrying a vertical load of 120 kN at an eccentricity of 50 mm in a plane bisecting the thickness as shown in Figure 2. Determine the maximum and minimum intensities of stress in the section and locate the position of the neutral axis.

(6 Marks)



- b) A steel beam with a symmetrical I-shaped cross-section with dimensions as given in Figure 3 is simply supported over a span of 6m and loaded as shown.
- Sketch the shear force and bending moment diagram for the beam and determine the maximum bending moment.

Hence, calculate:

$$b = 200 - a$$

$$\frac{10}{a} = \frac{-2}{200 - a}$$

$$2000 - 10a = -2a$$

$$2000 = 12a$$

$$a = 166.67$$

- ii). The maximum bending stress in the beam.
 iii). The radius of curvature of the beam, given $E = 205 \text{ kN/m}^2$.
 iv). The bending stress at the top of the web in the beam at the location of maximum bending moment. (14 Marks)

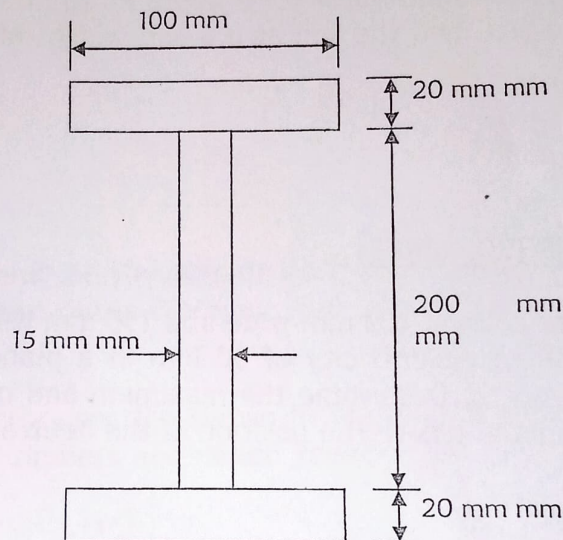
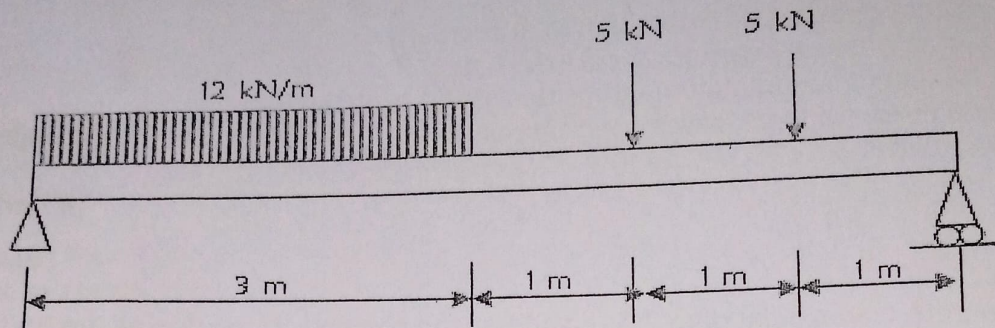


Figure 3

QUESTION FIVE (20 MARKS)

- a) Figure 4 shows a rolled steel beam of an unsymmetrical I section. Determine the centroid of the section. If the maximum bending stress in the beam section is not to exceed 50 MPa , find the moment which the beam can resist. (8 Marks)

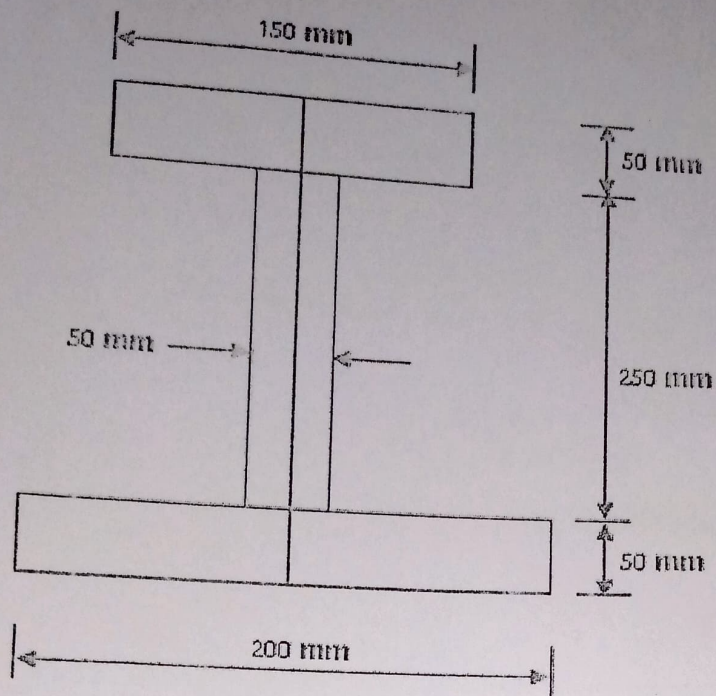


Figure 4

- b) A rectangular timber beam section is 50 mm wide and 100 mm deep. It is reinforced with a steel plate 20 mm thick at the bottom as shown in Figure 5 and loaded with a moment of 10 kNm. If E for timber is 11 GPa and for steel is 205 GPa, calculate the maximum stress in:

- The steel
- The timber

Sketch the true stress distribution diagram

(12 Marks)

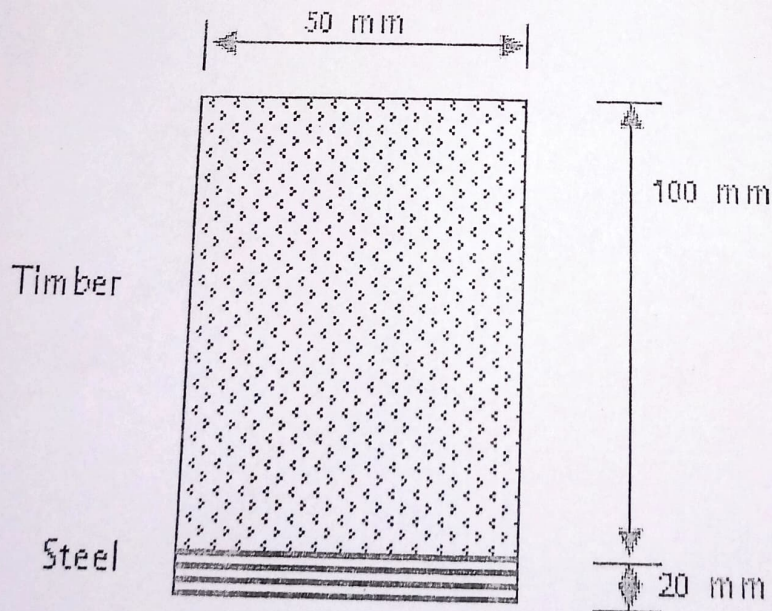


Figure 5